

FIN Research Monograph — Release 10.9

FIN Programs 81–90

Asymptotic Spectral Obstructions
and Operational Completion Tests

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: [0009-0002-0909-3613](https://orcid.org/0009-0002-0909-3613)

27 July 2026

Publication — Preprint; record version 1.0.0

Scope. Ten executed studies of harmonic-alias Schur reduction, large- N spectral obstructions, dense-kernel norm artefacts, fixed symbols, approximate locality, quotient actions, robust calibration, chiral-state dynamics, feedback thermodynamics, process tensors, and external-data admission.

Central correction. Frobenius convergence of dense normalized rows does not imply uniform symbol convergence or a continuum fixed point.

Guardrail. No strict selector, physical-unit source, Lorentz closure, legacy role transfer, completed legacy–strict bridge, or external physical validation is claimed.

License. Creative Commons Attribution 4.0 International (CC BY 4.0).

Publication statement

This release reports Programs 81–90 and recommends Programs 91–102. Theorem-level statements, numerical evidence, conditioned operational models, failed inferences, and open obligations are labeled separately.

Contents

Publication statement	i
0.1 Confidence convention	1
1 Executive summary	2
2 Mathematical setup	4
2.1 Kernels and declared positive realization	4
2.2 Four levels of interpretation	4
3 Program 81 — Asymptotic Fourier-symbol audit	5
3.1 Exact alias theorem	5
3.2 Three non-equivalent errors	5
3.3 Strict lattice-distance results	6
3.4 Coordinate-rescaled results	6
3.5 Verdict	6
4 Program 82 — Fixed-symbol classification	8
4.1 Exact affine-cosine cone	8
4.2 Wider-family experiment	8
4.3 Interpretation	9
4.4 Verdict	9
5 Program 83 — Dense-kernel universality and the norm artefact	10
5.1 Dilution theorem	10
5.2 Six-profile test	10
5.3 Exact constant-profile counterexample	11
5.4 Verdict	12
6 Program 84 — Approximate locality bounds	13
6.1 Theorem	13
6.2 C_{64} results	13

6.3	Verdict	14
7	Program 85 — Quotient action and entropy	15
7.1	Regulator-free action	15
7.2	Quotient partition function	15
7.3	Gradient flow	16
7.4	Verification	16
7.5	Verdict	16
8	Program 86 — Robust calibration with nuisance parameters	17
8.1	Model	17
8.2	Result	17
8.3	Verdict	18
9	Program 87 — Chiral-state generation and stability	19
9.1	State family	19
9.2	Reflection-symmetric dynamics	19
9.3	Explicit bias	19
9.4	Verdict	20
10	Program 88 — Feedback information thermodynamics	21
10.1	Protocol	21
10.2	Exact identities	21
10.3	Apparatus bookkeeping	22
10.4	Verdict	22
11	Program 89 — Two-slot process tensor	23
11.1	Operational map	23
11.2	Results	23
11.3	Meaning	23
11.4	Verdict	24
12	Program 90 — External-data admission gate	25
12.1	Required fields	25
12.2	Intake result	25
12.3	Synthetic runner check	25
12.4	Verdict	26
13	Integrated scientific consequences	27

13.1	Corrected continuum statement	27
13.2	Corrected renormalization statement	27
13.3	Corrected selector statement	27
13.4	Corrected information-to-physics statement	28
14	Current theorem and obstruction table	29
15	Recommended Programs 91–102	30
15.1	Ranked roadmap	30
15.2	Program 91 — Analytic strict-lattice nonclosure theorem	30
15.3	Program 92 — Critical tuning and field rescaling	31
15.4	Program 93 — Graphon and nonlocal continuum limit	31
15.5	Program 94 — Sharp long-range propagation theorem	31
15.6	Program 95 — Adaptive quotient gradient flow	32
15.7	Program 96 — Strict chiral-state source acceptance test	32
15.8	Program 97 — Noisy process-tensor witness optimization	32
15.9	Program 98 — Complete feedback apparatus cycle	32
15.10	Program 99 — External dataset acquisition	33
15.11	Program 100 — Typed completion-atom test	33
15.12	Program 101 — Fractional/integral operator comparison	33
15.13	Program 102 — Pure mathematical consolidation	33
16	Stop rules for the next cycle	35
17	Final judgment	36
18	Reproducibility	37
18.1	Suggested citation	37
	Archival note	38

0.1 Confidence convention

- **[Proven]** analytic theorem or exact finite algebra under the stated definitions.
- **[Strong evidence]** reproducible computation with a large margin and a frozen protocol.
- **[Moderate evidence]** a stable numerical pattern within the tested model class.
- **[Conditional]** requires a declared repair, normalization, scale, state law, bath, preparation, instrument, or calibration.
- **[Refuted]** a specific claim or map contradicted inside its declared scope.
- **[Open]** neither proved nor rigorously falsified.

The project ontology is retained as a premise: the nadsoliton is primordial fractal information in a solitonic state, with no separate informational layer underneath it. This premise is not promoted to experimentally established physics.

The strict kernel remains the primary strict working kernel. The legacy kernel remains an intermediate bridge object. Positive absolute-value realizations used below are named mathematical repairs; they do not prove the legacy-to-strict completion map and do not authorize physical-role transfer.

No result in this release closes QW-2191, derives an absolute unit, proves Lorentz symmetry, supplies a strict chiral state source, completes the legacy-to-strict bridge, transfers legacy physical roles, constructs a unit-bearing L_{total} , or provides external experimental evidence.

Chapter 1

Executive summary

Programs 81–90 test the strongest research directions left by Release 10.8. The main goal is to decide whether the apparent large- N projective convergence is genuine or an artefact of the norm, and then to determine how far explicit state, thermodynamic, calibration, and observer structures can be developed without falsely promoting them to strict FIN consequences.

The principal conclusion is:

The previously observed $N^{-1/2}$ Frobenius convergence is not a continuum theorem.

It is a generic dense-row dilution effect compatible with a nonzero spectral obstruction.

The strongest results are:

1. **Alternating-site Schur reduction has an exact harmonic-alias formula.** [Proven]

$$\lambda_{\text{coarse}}(k) = \frac{2\lambda(k)\lambda(k+N)}{\lambda(k) + \lambda(k+N)}.$$

Its numerical residual against direct block Schur reduction is

$$2.46 \times 10^{-16}.$$

2. **Strict lattice-distance projective closure is strongly contradicted.** [Strong evidence]

At $N = 49152$, the operator Frobenius defect is still 0.074284, the uniform symbol defect is 0.154303, and the relative defect of the first nonzero mode is 0.891376. None approaches zero in the tested range.

3. **Coordinate-rescaled Frobenius convergence is norm dilution, not uniform spectral convergence.** [Proven mechanism; strong numerical evidence]

For strict $K(d/N)$, the Frobenius defect falls to

$$6.181 \times 10^{-4}$$

with slope -0.5008 , while the uniform symbol defect remains

$$0.132518.$$

4. **The $N^{-1/2}$ law is generic across unrelated dense profiles.** [Proven bound and six-profile test]

Constant, exponential, Gaussian, rational-power, strict, and legacy profiles all have Frobenius slopes between -0.503 and -0.501 , while every final uniform symbol defect exceeds 0.13.

5. **A constant-profile counterexample is exact.** [Proven]

With fixed mass $m = 1/4$, the normalized native zero mode tends to

$$\frac{m}{m+1} = \frac{1}{5},$$

whereas the normalized Schur zero mode tends to

$$\frac{2m}{2m+1} = \frac{1}{3}.$$

Therefore the limiting uniform gap is exactly

$$\frac{2}{15} = 0.133333\dots$$

even though the row-Frobenius defect vanishes.

6. **The fixed positive mass drives normalized Schur flow toward an ultralocal constant symbol.** [Proven in the affine-cosine cone; strong wider-family evidence]

The massless nearest-neighbour symbol is a normalized fixed point, but any positive mass obeys $r' = r(r+4)$ and flows toward the constant-symbol boundary. The repaired strict precision's distance to the constant symbol decreases from 0.29910 to 3.79×10^{-4} in eight steps.

7. **Approximate locality admits a rigorous but loose finite-volume bound.** [Proven]

A radius- R truncation gives a factorial series tail plus a Duhamel error $t\|L - L_R\|$. For the opposite sites of C_{64} , the best amplitude bounds are 0.01472, 0.08532, and 0.17919 at $t = 0.1, 0.5, 1$, respectively. These are not Lorentz cones.

8. **The quadratic action and Gaussian entropy are regulator-free on $1^\perp$.** [Proven]

Gauge invariance, pseudodeterminant partition functions, unique quotient stationarity, and monotone Dirichlet energy were verified with residuals from 10^{-15} to 10^{-10} .

9. **Calibration remains possible under explicit nuisance parameters, but requires redundant external standards.** [Conditional strong result]

A maximin D-optimal design over 118755 allocations assigns 12, 6, 12, 8, 12, 10 records to six measurement classes. Omitting both length classes, both clock classes, or the energy class reduces rank from five to four.

10. **Spontaneous chiral branching does not solve the selector.** [Proven]

Reflection-symmetric nonlinear state dynamics has paired stable branches $q = \pm 1$. Symmetric perturbations choose them 49.52%/50.48%. A unique positive branch appears only after an explicit bias $h = 1.2$ is inserted.

11. **Feedback thermodynamics closes exactly only after measurement and apparatus semantics are supplied.** [Proven conditional identity]

The binary feedback model saturates

$$\langle W \rangle = \Delta F - I$$

and its generalized Jarzynski equality to machine precision. Resetting the apparatus memory restores the complete information cost.

12. **A two-slot process witness is stronger than a one-time spectrum.** [Strong finite result]

Nonselective dephasing changes the wave branch but leaves classical diffusion unchanged. At half-time 0.5, the wave intervention contrast is 0.085392, while the diffusion contrast is exactly zero.

13. **No admissible external dataset is currently present.** [Proven intake result]

The intake directory contains no candidate manifest. A synthetic strict-versus-nearest-neighbour dry run validates the direction of the scoring code, but supplies no physical evidence.

Chapter 2

Mathematical setup

2.1 Kernels and declared positive realization

The frozen kernels are

$$K_{\text{legacy}}(d) = 4 \ln 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d},$$

$$K_{\text{strict}}(d) = \frac{\cos(0.18575d + 0.16250)}{1 + d^{1.8}}.$$

On C_N , whenever a positive precision is required, the declared weights are

$$w_{ij} = \frac{|K(d_N(i, j))|}{\sum_{k \neq i} |K(d_N(i, k))|}, \quad w_{ii} = 0,$$

and

$$A_N = mI + L(w), \quad m = \frac{1}{4}.$$

The fixed mass is a numerical regularizer and scale convention. It is not a derived physical mass.

2.2 Four levels of interpretation

This release keeps separate:

1. exact finite operator identities;
2. repaired or normalized mathematical models;
3. operationally conditioned models containing states, instruments, baths, clocks, and calibration;
4. comparison with independent experimental records.

Programs 81–89 reach levels one through three. Program 90 tests admission to level four and finds no admitted dataset.

Chapter 3

Program 81 — Asymptotic Fourier-symbol audit

3.1 Exact alias theorem

Let A be a positive circulant precision on C_{2N} , with eigenvalues

$$\lambda_0, \dots, \lambda_{2N-1}.$$

Retaining the even sites aliases the two fine Fourier modes k and $k + N$. The retained Green eigenvalue is their arithmetic mean:

$$g_{\text{ret}}(k) = \frac{1}{2} \left(\frac{1}{\lambda_k} + \frac{1}{\lambda_{k+N}} \right).$$

The retained precision is its inverse:

$$\lambda_{\text{Schur}}(k) = \frac{2\lambda_k \lambda_{k+N}}{\lambda_k + \lambda_{k+N}}.$$

Thus Schur reduction is a harmonic mean across each alias pair.

The relative residual against the two-sublattice block formula is

$$2.46 \times 10^{-16}.$$

3.2 Three non-equivalent errors

For normalized compressed and native operators, the study records:

$$\varepsilon_F = \frac{\|A_N^{\text{Schur}} - A_N^{\text{native}}\|_F}{\|A_N^{\text{native}}\|_F},$$
$$\varepsilon_\infty = \frac{\max_k |\lambda_k^{\text{Schur}} - \lambda_k^{\text{native}}|}{\max_k |\lambda_k^{\text{native}}|},$$

and the relative defect of the first nonzero mode.

3.3 Strict lattice-distance results

retained N	Frobenius defect	uniform defect	symbol	mode-1 defect	Green defect	Frobenius
192	0.075654	0.150760		0.737859	0.273524	
768	0.074695	0.153238		0.834504	0.261337	
3072	0.074407	0.154256		0.873221	0.257606	
12288	0.074314	0.154317		0.886808	0.256408	
49152	0.074284	0.154303		0.891376	0.256016	

The fitted Frobenius slope is only -0.0030 . The mode-1 error increases.

3.4 Coordinate-rescaled results

retained N	Frobenius defect	uniform defect	symbol	mode-1 defect	Green defect	Frobenius
192	0.009941	0.132495		0.022197	0.136968	
768	0.004951	0.132512		0.022867	0.071196	
3072	0.002473	0.132517		0.023034	0.035965	
12288	0.001236	0.132518		0.023075	0.018029	
49152	0.000618	0.132518		0.023086	0.009020	

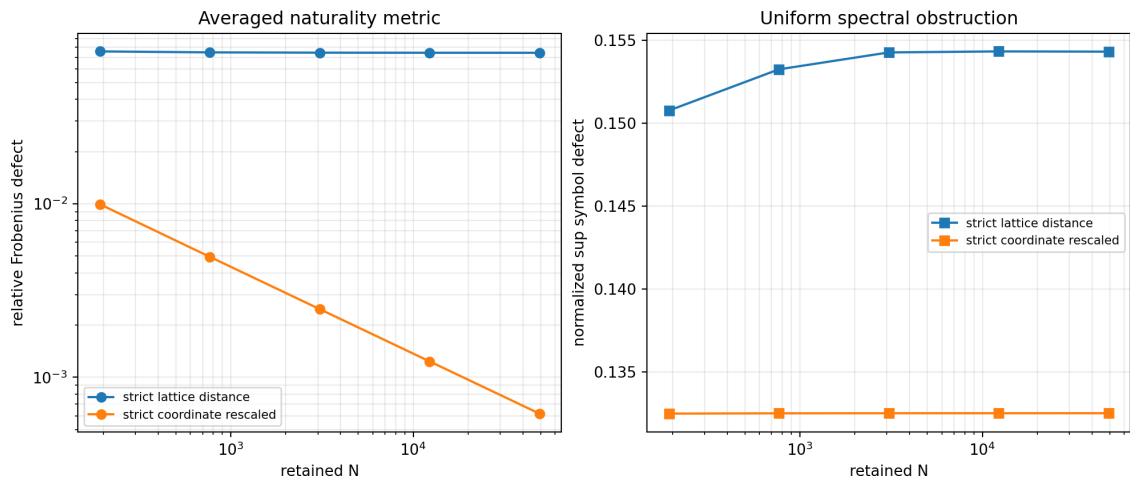


Figure 3.1: Frobenius and uniform-symbol naturality metrics through retained size 49152.

3.5 Verdict

[Strong evidence] The strict lattice-distance family is not approaching native Schur-projective closure under the frozen map.

[Refuted] Vanishing Frobenius error does not imply uniform symbol convergence for the coordinate-rescaled family.

[Open] A strict-specific analytic positive lower bound remains to be proved. The numerical evidence identifies its likely value but is not itself the theorem.

Chapter 4

Program 82 — Fixed-symbol classification

4.1 Exact affine-cosine cone

Consider

$$\lambda(q) = m + 2c(1 - \cos q), \quad m \geq 0, \quad c > 0.$$

The harmonic-alias map preserves this family and gives

$$c' = \frac{c^2}{m + 2c}, \quad m' = \frac{m(m + 4c)}{m + 2c}.$$

For $r = m/c$,

$$r' = r(r + 4).$$

After normalization by the mean symbol:

- $r = 0$ is the massless nearest-neighbour fixed symbol;
- $r = \infty$ is the ultralocal constant-symbol boundary;
- every finite $r > 0$ moves away from the massless point and toward the constant boundary.

4.2 Wider-family experiment

Family	initial distance to constant	distance after 8 steps
constant	0	0
massless nearest-neighbour	0.707107	0.707107
massive nearest-neighbour, $r = 0.1$	0.673435	1.18×10^{-16}
strict positive precision	0.299102	3.79×10^{-4}
legacy absolute precision	0.019599	0.003805

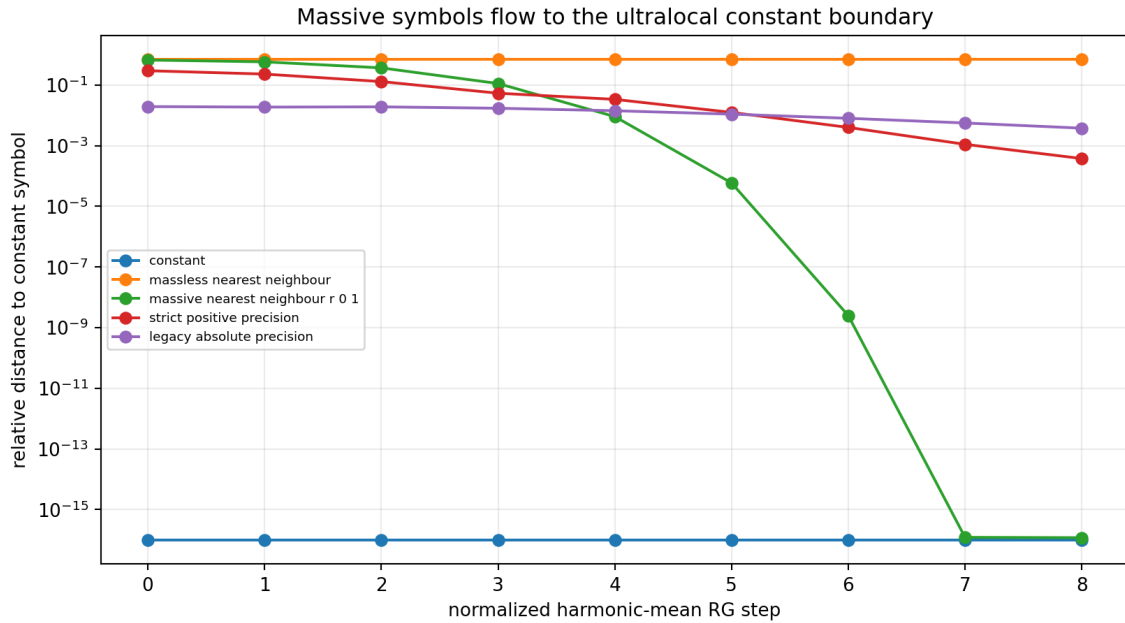


Figure 4.1: Normalized harmonic-alias flow toward the constant-symbol boundary for positive-mass families.

4.3 Interpretation

The fixed regularizer is not passive under blocking. It is a relevant direction. Repeated normalized compression erases spatial structure and approaches an ultralocal symbol unless the massless condition is imposed and its scale is renormalized appropriately.

This explains why a compression semigroup can be exact while failing to produce a nontrivial continuum field theory.

4.4 Verdict

[Proven] The affine-cosine fixed-point classification follows from the exact ratio map.

[Strong evidence] The repaired strict and legacy positive symbols also flow toward the constant boundary under the tested normalized map.

[Conditional] The conclusion concerns the declared mass, repair, and normalization. It is not a theorem about the raw signed legacy kernel.

Chapter 5

Program 83 — Dense-kernel universality and the norm artefact

5.1 Dilution theorem

Let two $N \times N$ circulant precisions have:

1. the same diagonal bounded below independently of N ;
2. off-diagonal row differences bounded by C/N ;
3. uniformly bounded inverse for the eliminated block.

Then Schur complementation preserves the $O(1/N)$ scale of dense off-diagonal rows, and

$$\|A_N - B_N\|_F = \sqrt{N} \|a_N - b_N\|_2 = O(1),$$

while

$$\|B_N\|_F = O(\sqrt{N}).$$

More directly at the row level,

$$\|a_N - b_N\|_2 = O(N^{-1/2}),$$

so the relative Frobenius naturality defect can vanish as $N^{-1/2}$ even when a finite set of Fourier modes retains an $O(1)$ mismatch.

5.2 Six-profile test

Coordinate profile	Frobenius slope	final uniform symbol defect
constant	−0.5022	0.133334
exponential	−0.5009	0.133323
Gaussian	−0.5028	0.141724
rational power	−0.5019	0.132987
strict	−0.5022	0.132517

Coordinate profile	Frobenius slope	final uniform symbol defect
legacy absolute	−0.5021	0.132924

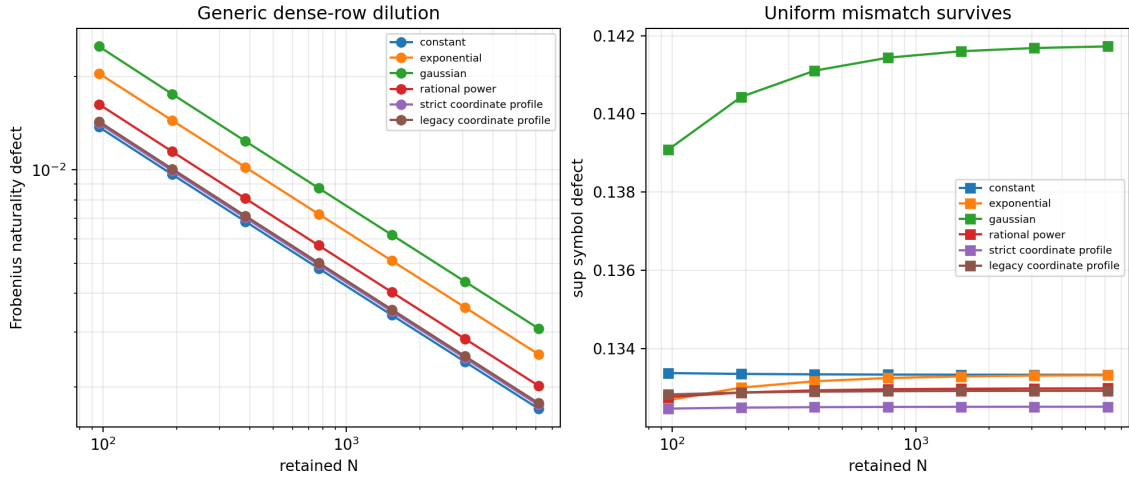


Figure 5.1: Generic dense-row Frobenius dilution alongside persistent uniform spectral defects.

5.3 Exact constant-profile counterexample

For the complete-graph profile and fixed mass m , the normalized native zero mode tends to

$$\mu_{\text{native}}(0) = \frac{m}{m+1}.$$

Harmonic aliasing of the constant mode with a bulk mode gives

$$\mu_{\text{Schur}}(0) = \frac{2m}{2m+1}.$$

At $m = 1/4$:

$$\mu_{\text{native}}(0) = \frac{1}{5}, \quad \mu_{\text{Schur}}(0) = \frac{1}{3},$$

and therefore

$$\|\mu_{\text{Schur}} - \mu_{\text{native}}\|_{\infty} \geq \frac{2}{15}.$$

This is an explicit theorem-level counterexample to the inference

$$\varepsilon_F(N) \rightarrow 0 \implies \text{uniform spectral closure.}$$

5.4 Verdict

[Proven] The $N^{-1/2}$ Frobenius rate is generic under dense normalized sampling.

[Refuted] The observed strict/legacy coordinate-rescaled rate is not evidence unique to FIN.

[Research correction] Continuum tests must use uniform symbol, resolvent, Mosco, or another topology that controls the physically relevant modes.

Chapter 6

Program 84 — Approximate locality bounds

6.1 Theorem

Let L_R be a radius- R truncation of a positive graph generator, and let m be the minimum number of allowed hops from x to y . Since

$$(L_R^k)_{yx} = 0 \quad \text{for } k < m,$$

the exponential series gives

$$\left| \left(e^{zL_R} \right)_{yx} \right| \leq \sum_{k=m}^{\infty} \frac{(|z| \|L_R\|)^k}{k!}.$$

For the full and truncated self-adjoint unitary groups,

$$\|e^{-itL} - e^{-itL_R}\| \leq |t| \|L - L_R\|.$$

The same Duhamel estimate holds for the positive contractive heat semigroups:

$$\|e^{-tL} - e^{-tL_R}\| \leq t \|L - L_R\|.$$

Therefore a valid full-kernel far-site bound is

$$B_R(t) = \sum_{k=m}^{\infty} \frac{(t \|L_R\|)^k}{k!} + t \|L - L_R\|.$$

6.2 C_{64} results

For source and target separated by 32 sites:

t	best radius	hop der	or- rigorous tude bound	ampli- actual wave ampli- tude	actual diffusion en- try
0.1	16	2	0.014724	0.00010283	0.00010251
0.5	8	4	0.085322	0.00051570	0.00050552
1.0	4	8	0.179187	0.00103969	0.00099158

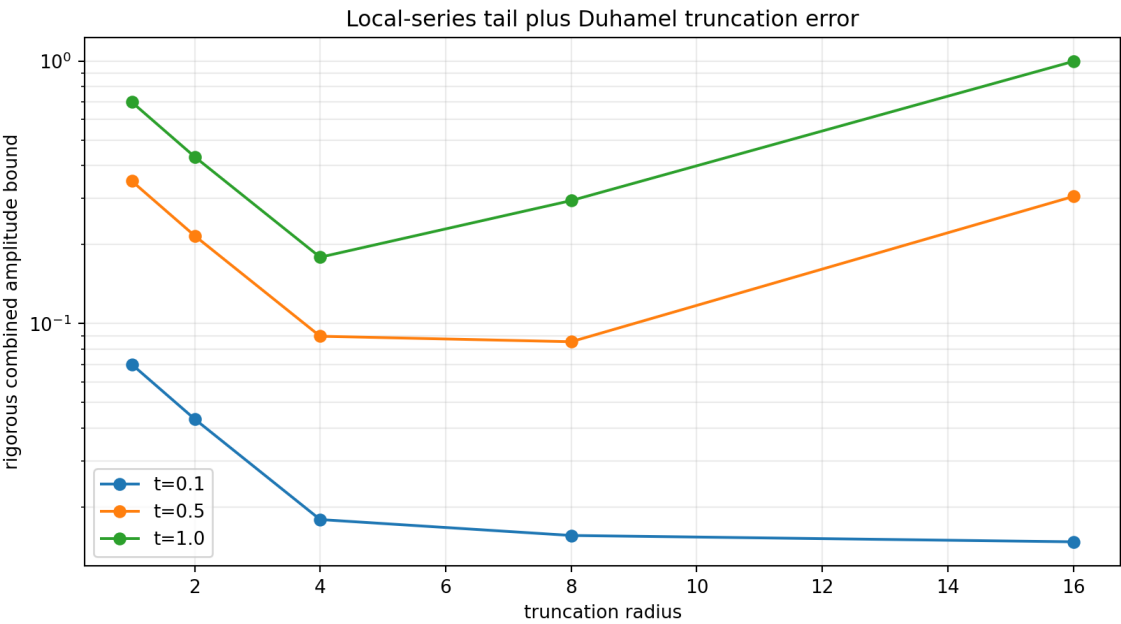


Figure 6.1: Finite-volume locality bounds combining a factorial series tail with Duhamel truncation error.

6.3 Verdict

[Proven] The combined factorial-tail and truncation estimate bounds every measured far-site value.

[Moderate usefulness] The bounds are nontrivial but much looser than the actual amplitudes.

[Not derived] Graph distance, interaction radius, and dimensionless time do not define a Lorentzian light cone or a physical propagation speed.

Chapter 7

Program 85 — Quotient action and entropy

7.1 Regulator-free action

Let L be a connected positive Laplacian and $J \perp \mathbf{1}$. Define

$$S[f] = \frac{1}{2} \langle f, Lf \rangle - \langle J, f \rangle.$$

Since $L\mathbf{1} = 0$,

$$S[f + c\mathbf{1}] = S[f].$$

The stationary class is

$$f_* = L^+ J + c\mathbf{1}.$$

It is unique on the quotient

$$\mathbb{R}^N / \text{span}\{\mathbf{1}\} \simeq \mathbf{1}^\perp.$$

7.2 Quotient partition function

With $L_0 = L|_{\mathbf{1}^\perp}$,

$$\log Z(J) = \frac{N-1}{2} \log(2\pi) - \frac{1}{2} \log \text{pdet } L + \frac{1}{2} \langle J, L^+ J \rangle.$$

The Gaussian differential entropy is

$$H = \frac{1}{2} [(N-1)(1 + \log 2\pi) - \log \text{pdet } L].$$

7.3 Gradient flow

For

$$\dot{f} = -Lf,$$

the Dirichlet energy

$$E[f] = \frac{1}{2}\langle f, Lf \rangle$$

satisfies

$$\frac{dE}{dt} = -\|Lf\|^2 \leq 0.$$

7.4 Verification

Realization	stationarity residual	residual	gauge residual	partition residual	derivative residual
strict positive	2.82×10^{-15}		9.33×10^{-15}	0	1.03×10^{-10}
legacy absolute	2.81×10^{-15}		1.78×10^{-15}	1.78×10^{-15}	1.24×10^{-11}

7.5 Verdict

[Proven] The quadratic action, Gaussian normalization, and dissipative energy law require no mass regulator after quotienting the constant mode.

[Conditional] The legacy computation still uses the named absolute-value repair.

[Not physical entropy] Gaussian differential entropy becomes thermodynamic entropy only after physical state, temperature, and measurement semantics are supplied.

Chapter 8

Program 86 — Robust calibration with nuisance parameters

8.1 Model

The parameter vector is

$$\theta = (\log \ell_*, \log \tau_*, \log \hbar_*, \delta_{\text{clock}}, \delta_{\text{distance}}).$$

Six record classes are allowed:

Class	Jacobian
primary length	(1, 0, 0, 0, 1)
length cross-check	(1, 0, 0, 0, 0)
primary clock	(0, 1, 0, 1, 0)
clock cross-check	(0, 1, 0, 0, 0)
energy standard	(0, -1, 1, 0, 0)
length-time link	(1, -1, 0, 0, 0)

The design maximizes the worst-case Fisher log determinant across nominal, length-degraded, clock-degraded, energy-degraded, and link-degraded noise scenarios.

8.2 Result

Among 118755 even allocations of 60 records, the maximin design is:

Record class	count
primary length	12
length cross-check	6
primary clock	12

Record class	count
clock cross-check	8
energy standard	12
length-time link	10

The worst scenario is degraded clock accuracy. Its Fisher condition number is 36.20. The worst-case one-sigma log-parameter bounds are

$$(0.00441, 0.00416, 0.00711, 0.00490, 0.00527).$$

Omitting both length classes, both clock classes, or the energy class reduces rank to four. Omitting only the link retains rank five because it is a redundancy rather than the sole source of a parameter direction.

8.3 Verdict

[Strong conditional result] The scale conversion package can be estimated robustly when redundant external standards are supplied.

[Proven obstruction] Dimensionless FIN records cannot distinguish an absolute scale from an unconstrained calibration bias.

Chapter 9

Program 87 — Chiral-state generation and stability

9.1 State family

Let ρ_+ and ρ_- be the $k = +1$ and $k = -1$ Fourier states. Define

$$\rho(q) = \frac{1+q}{2}\rho_+ + \frac{1-q}{2}\rho_-, \quad q \in [-1, 1].$$

The strict state-current functional is linear:

$$\Lambda(\rho(q), W_{\text{strict}}) = q \Lambda(\rho_+, W_{\text{strict}}),$$

with

$$\Lambda(\rho_+, W_{\text{strict}}) = -0.4699856726.$$

9.2 Reflection-symmetric dynamics

The bounded nonlinear law

$$\dot{q} = (1 - q^2)q$$

has stable fixed points $q = \pm 1$ and unstable fixed point $q = 0$. Reflection maps $q \mapsto -q$.

Initial conditions $\pm 10^{-3}$ converge to the opposite stable branches with equal magnitude. In 20000 symmetric Gaussian perturbations, the positive branch frequency is 0.4952, statistically consistent with one half.

9.3 Explicit bias

Adding

$$\dot{q} = (1 - q^2)(q + h)$$

with $h = 1.2$ makes $q = +1$ the only stable branch reachable from the interior. All 1000 tested perturbations converge positively, with mean final value

0.99999999997.

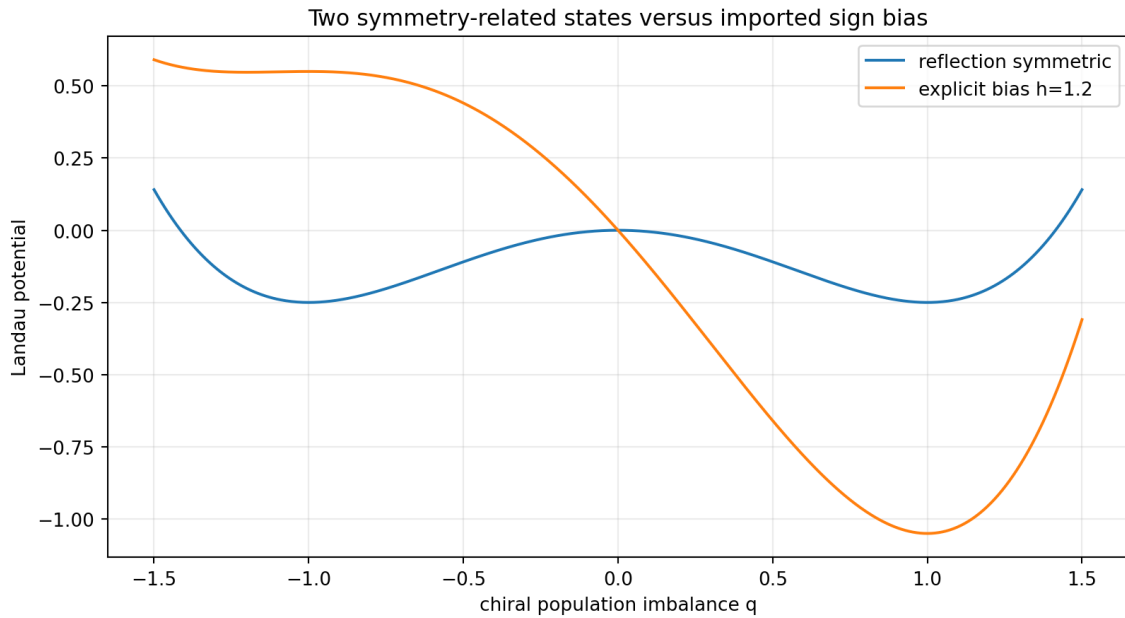


Figure 9.1: Paired reflection-symmetric chiral sectors and an explicitly biased state law.

9.4 Verdict

[Proven] Reflection-equivariant deterministic dynamics cannot distinguish the paired attractors.

[Proven] A strong signed bias can select one branch.

[Not strict] The nonlinear state law and bias are additional assumptions. Spontaneous branching produces two possible sectors, not a canonical selector. QW-2191 remains open.

Chapter 10

Program 88 — Feedback information thermodynamics

10.1 Protocol

Let $X \in \{0, 1\}$ be equiprobable. A binary measurement Y has error probability ϵ :

$$\Pr(Y \neq X) = \epsilon.$$

The mutual information is

$$I(X : Y) = \ln 2 - h(\epsilon),$$

where

$$h(\epsilon) = -(1 - \epsilon) \ln(1 - \epsilon) - \epsilon \ln \epsilon.$$

After observing Y , the protocol raises the energy of the unobserved alternative by

$$\Delta = \ln \frac{1 - \epsilon}{\epsilon}.$$

The posterior distribution is then exactly Gibbs for the feedback Hamiltonian.

10.2 Exact identities

The mean work is

$$\langle W \rangle = \epsilon \Delta.$$

The free-energy difference obeys

$$\langle W \rangle = \Delta F - I.$$

Trajectory information satisfies

$$\langle \exp[-W + \Delta F - i(X, Y)] \rangle = 1.$$

All tested residuals for $\epsilon = 0.01, 0.05, 0.1, 0.2, 0.4$ are below 1.2×10^{-16} .

At $\epsilon = 0.1$:

$$I = 0.3680642, \quad \Delta F = 0.5877867, \quad \langle W \rangle = 0.2197225.$$

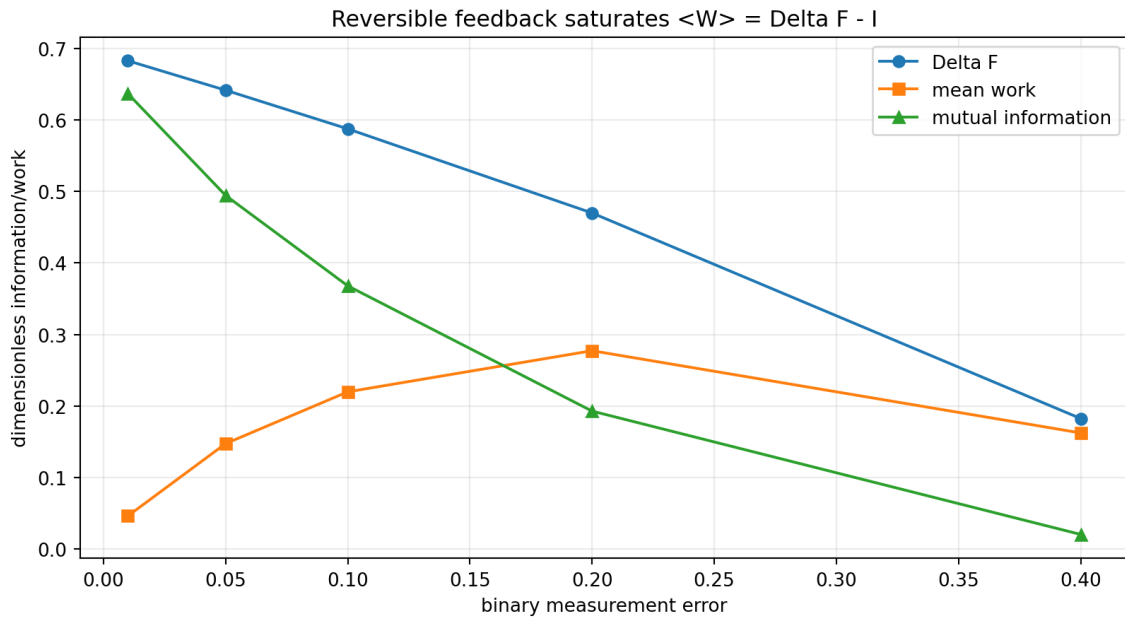


Figure 10.1: Feedback work, free-energy change, and mutual information versus binary measurement error.

10.3 Apparatus bookkeeping

The memory has entropy $\ln 2$. Conditional reset with access to X costs $h(\epsilon)$, and the saving is exactly

$$\ln 2 - h(\epsilon) = I.$$

Thus feedback does not turn information into free energy without accounting for preparation and reset of the measurement memory.

10.4 Verdict

[Proven conditional identity] The feedback second law and generalized Jarzynski equality close exactly in the specified model.

[Not derived] Measurement channel, bath, Hamiltonian, temperature, feedback controller, and memory reset are operational inputs, not consequences of the FIN kernel.

Chapter 11

Program 89 — Two-slot process tensor

11.1 Operational map

For an intermediate operation $\mathcal{M}$, define

$$\mathcal{T}[\mathcal{M}] = \mathcal{E}_{t_2} \circ \mathcal{M} \circ \mathcal{E}_{t_1}(\rho_0).$$

The tested interventions are:

1. identity;
2. nonselective dephasing in the site basis;
3. a recorded site instrument producing a joint intermediate/final law.

The wave branch uses unitary density-matrix evolution. The diffusion branch uses the classical Markov semigroup.

11.2 Results

Half-time	wave dephasing contrast	diffusion contrast	joint-record JSD
0.10	0.003883	0	0.058888
0.25	0.023601	0	0.120622
0.50	0.085392	0	0.178597

Every joint law is normalized to residual below 2.3×10^{-16} .

11.3 Meaning

Dephasing destroys phase coherence in the wave process but is operationally invisible to an already classical probability vector. This intervention sensitivity is not encoded by the final-time spectrum alone.

The result gives a concrete answer to the observer question:

$$\text{same generator} \not\Rightarrow \text{same process tensor}.$$

Preparations and interventions determine which functional calculus is tested.

11.4 Verdict

[Strong finite result] A minimal two-slot experiment distinguishes wave coherence from classical diffusion.

[Conditional] The intervention, site basis, clock, and final instrument are external operational structure.

Chapter 12

Program 90 — External-data admission gate

12.1 Required fields

An external dataset must document:

- independent producer;
- preparation map;
- instrument or POVM;
- clock calibration;
- length calibration;
- environment and boundary conditions;
- raw counts;
- license;
- frozen train/validation/test assignment.

Every field must be non-null before unblinding.

12.2 Intake result

No `external_data` intake directory is present. Consequently:

$$\text{candidate manifests} = 0, \quad \text{admitted datasets} = 0.$$

The generated template is cryptographically frozen with canonical digest:

7c533e7433fc2cfb34b3ce6f25068e9cd2ceaa3b73333d97ca67a31a6ddd7b98

12.3 Synthetic runner check

As a code-directionality test only, 300 strict-generated and 300 nearest-neighbour-generated trials were scored using the two frozen candidates. Each truth was selected correctly in all trials. Minimum log-likelihood margins were 37.77 and 47.56, respectively.

This demonstrates that the runner can both support strict when strict generated the records and reject strict when the null generated them. It is not evidence about nature.

12.4 Verdict

[**Proven**] No admissible external-data claim can currently be made.

[**Strong methodology check**] The frozen runner passes a symmetric synthetic directionality test.

[**Guardrail**] Synthetic power and type-I-error checks must never be reported as experimental validation.

Chapter 13

Integrated scientific consequences

13.1 Corrected continuum statement

The valid chain is:

exact finite Schur reduction $\implies$ exact harmonic aliasing.

It does not imply:

native family closure or uniform symbol convergence,
a local continuum field or Lorentz geometry.

The coordinate-rescaled family may have a meaningful nonlocal dense-operator or graphon limit, but the Frobenius slope alone cannot identify it.

13.2 Corrected renormalization statement

The fixed positive mass is a relevant RG direction. Under the tested normalization, massive positive symbols flow toward an ultralocal constant generator. A nontrivial continuum theory would require a justified mass tuning, field rescaling, or another topology.

Such tuning cannot be performed post hoc merely to restore a desired fixed point.

13.3 Corrected selector statement

The kernel-state current supplies the correct representation for chirality. Nonlinear dynamics can stabilize two chiral sectors. However:

paired stable sectors $\neq$ unique strict selector.

A signed bias selects a sector only by adding the missing symmetry-breaking datum.

13.4 Corrected information-to-physics statement

Quotient actions, Gaussian entropy, feedback work, and process tensors are mathematically coherent. Their physical meanings require:

$$\mathfrak{D} = (\rho, \mathcal{E}_t, \mathcal{P}, \mathcal{I}, \mathcal{A}, \mathcal{R}, \mathcal{C}),$$

plus a bath and Hamiltonian for thermodynamics.

The operational package is not decorative. It is the object that turns an operator identity into a falsifiable record law.

Chapter 14

Current theorem and obstruction table

Target	Best result after Program 90	Remaining obligation	Status
alternating-site reduction	exact harmonic-alias theorem	none in finite scope	Proven
strict lattice continuum	nonzero defects through $N = 49152$	analytic positive bound	Strong no-go evidence
coordinate continuum	generic $N^{-1/2}$ dilution	correct topology and limiting operator	Frobenius inference refuted
RG fixed point	massless NN and constant boundary classified	justified tuning/ rescaling for FIN	Open
locality	factorial plus Duhamel bound	sharp long-range theorem and CA	Conditional mathematical
variational action	quotient action and pdet partition	physical field/state semantics	Proven mathematical
dimensions	robust rank-five CA design	external standards	Conditional
selector	paired stable chiral sectors	strict signed source	Open
information thermodynamics	exact feedback identities	physical apparatus	bath/ Conditional
observer	two-slot process witness	experimental implementation	Conditional
physical evidence	frozen admission template	independent admissible data	Absent

Chapter 15

Recommended Programs 91–102

Probabilities estimate the chance of obtaining a decisive theorem, obstruction, or reproducible scientific result. They do not estimate the probability that FIN is physically correct.

15.1 Ranked roadmap

Rank	Program	Main objective	Success probability
1	102	consolidate exact results into a pure mathematical paper	0.92
2	91	prove a strict-lattice positive symbol-defect lower bound	0.86
3	93	identify the graphon/nonlocal continuum limit	0.82
4	94	derive sharp long-range propagation bounds for $\eta = 1.8$	0.78
5	92	classify mass tuning and field rescaling	0.76
6	95	adaptive gradient flow on the quotient	0.72
7	97	optimize a noisy two-slot process witness	0.70
8	101	compare the strict limit with fractional and integral operators	0.68
9	98	full apparatus-inclusive feedback cycle	0.62
10	96	strict chiral state-source acceptance test	0.50
11	99	acquire and preregister an independent dataset	0.40
12	100	test one genuinely new legacy-to-strict completion atom	0.30

15.2 Program 91 — Analytic strict-lattice nonclosure theorem

Objective. Replace the $N = 49152$ evidence by a theorem.

Method.

1. derive the limiting lattice-distance Fourier series of the normalized strict row;

2. apply the harmonic-alias map analytically;
3. isolate the first mode and the maximizing symbol mode;
4. prove either

$$\liminf_{N \rightarrow \infty} \varepsilon_{\infty}(N) > 0$$

or an explicit counter-rate;

5. propagate the bound to the Green/resolvent topology.

Expected outcome. A scoped nonclosure theorem for the frozen repair, mass, and normalization.

15.3 Program 92 — Critical tuning and field rescaling

Objective. Determine whether a nontrivial fixed symbol is available only after a mathematically justified critical tuning.

Method.

- allow m_N and a field factor Z_N ;
- derive their exact transformation under harmonic aliasing;
- classify scaling laws that preserve a nonconstant low-frequency symbol;
- forbid target-fitted choices;
- distinguish massless critical limits from imported physical masses.

Stop rule. If every admissible target-independent scaling flows to the constant or singular boundary, issue a no-go theorem.

15.4 Program 93 — Graphon and nonlocal continuum limit

Objective. Identify what $K(d/N)$ actually converges to.

Method.

1. construct the integral operator on the unit circle with kernel $g(d(x, y))$;
2. test graphon cut convergence, Hilbert–Schmidt convergence, and resolvent convergence separately;
3. derive the continuum Schur/conditional-covariance operation;
4. compare its zero mode with the exact 2/15 obstruction;
5. determine whether the correct limit is nonlocal rather than field-theoretic.

15.5 Program 94 — Sharp long-range propagation theorem

Objective. Replace the loose finite truncation bound by a decay-sensitive theorem.

Method.

- use the strict $d^{-1.8}$ -type envelope;
- derive polynomial Lieb–Robinson or convolution bounds appropriate to one-dimensional long-range interactions;
- treat unitary amplitude and heat-kernel probability separately;
- quantify finite-cycle wrap-around;
- retain dimensionless graph distance until CA is supplied.

15.6 Program 95 — Adaptive quotient gradient flow

Objective. Determine whether the adaptive FIN learning law is a gradient flow of a regulator-free quotient functional.

Method.

1. project all state variables to $\mathbf{1}^\perp$;
2. identify the required metric on operator space;
3. test integrability of the adaptive vector field;
4. compute curl/Helmholtz obstructions when no potential exists;
5. prove monotonicity or exhibit a finite cycle.

15.7 Program 96 — Strict chiral-state source acceptance test

Objective. Test one explicit formula

$$\text{Source}_\chi(\text{nadsoliton}) \longmapsto \rho_\chi$$

against:

- nonzero signed current;
- reflection covariance;
- translation/origin independence;
- nonconventional sign provenance;
- dynamical stability;
- coupling to the existing orientation torsor.

Without a new formula-level source, do not replay generic chiral candidate inventories.

15.8 Program 97 — Noisy process-tensor witness optimization

Objective. Minimize the experimental resources needed to distinguish wave coherence from diffusion.

Method.

- optimize intervention time, dephasing strength, preparation set, and coarse POVM;
- add detector error and clock jitter;
- compute Fisher/Chernoff bounds;
- preregister a finite-shot decision threshold;
- verify type-I error against classical null processes.

15.9 Program 98 — Complete feedback apparatus cycle

Objective. Include measurement, control, memory, reset, and bath in one energy/information ledger.

Method.

- use a finite-time master equation;
- record trajectory work, heat, mutual information, and memory entropy;

- verify integral fluctuation relations;
- test imperfect reset and feedback delay;
- identify which costs remain when the system returns to its initial state.

15.10 Program 99 — External dataset acquisition

Objective. Populate the frozen Program-90 intake without weakening its requirements.

Requirements.

- the producer is independent of FIN;
- raw records and full operational provenance are available;
- calibration is independent;
- candidate models and nulls remain frozen;
- no kernel parameter is changed after inspection.

If no scientifically compatible dataset exists, report that obstruction rather than using a semantically unrelated dataset.

15.11 Program 100 — Typed completion-atom test

Objective. Reopen legacy-to-strict work only for one genuinely new object, such as:

- a target-independent compression exponent source;
- a strict phase/topological completion law;
- a noncircular positive damping source.

The candidate must pass provenance, target independence, uniqueness, symbol prediction, transfer across sizes, and only then a separate physical-role audit.

15.12 Program 101 — Fractional/integral operator comparison

Objective. Determine whether the nonlocal strict continuum candidate is equivalent to a known integral or fractional operator.

Method.

- compare low-frequency symbols;
- compare kernel singularity and tail class;
- test norm-resolvent equivalence;
- distinguish finite graphon kernels from fractional Laplacians;
- quantify mathematical novelty only after an equivalence theorem.

15.13 Program 102 — Pure mathematical consolidation

Objective. Publish the durable content independently of cosmology.

The paper should contain:

- harmonic-alias Schur theorem;
- exact massive nearest-neighbour RG;
- dense-row dilution theorem and constant-profile counterexample;

-
- strict-lattice numerical obstruction with a clearly stated theorem target;
 - quotient action and Gaussian information geometry;
 - approximate locality theorem;
 - operational distinction between unitary and Markov process tensors.
-

Chapter 16

Stop rules for the next cycle

1. Do not continue Frobenius-only continuum scans.
 2. Do not tune m_N or Z_N to the strict target after seeing the target.
 3. Do not call a graphon/nonlocal limit a local spacetime.
 4. Do not call paired chiral attractors a selector.
 5. Do not treat feedback information as free energy after omitting apparatus reset.
 6. Do not use synthetic records as external evidence.
 7. Do not reopen the bridge without one new typed completion atom.
-

Chapter 17

Final judgment

Programs 81–90 materially change the continuum assessment. The apparent coordinate-rescaled convergence found earlier is real in an averaged Frobenius sense but scientifically weaker than it appeared. The exact constant-profile counterexample proves that this norm can converge while a finite uniform spectral obstruction survives.

The finite theory itself becomes clearer:

Schur compression is harmonic aliasing of precision spectra.

With a fixed positive mass, the normalized flow tends toward an ultralocal constant symbol. The massless nearest-neighbour symbol is a critical fixed point, not the generic endpoint.

The operational theory also becomes clearer. Chirality requires a state law; thermodynamic advantage requires a measurement-and-memory ledger; the observer problem requires a process tensor; physical claims require independently calibrated data.

The deepest interpretation that survives this round is:

FIN is a finite spectral information-reduction framework.

It has a plausible nonlocal continuum direction, but:

No local physical continuum follows from the current kernel.

Nor does the current kernel select a unique operational sector.

The next decisive mathematical work is Program 91 or 93: prove the strict-lattice nonclosure bound, or identify the correct nonlocal continuum topology. The next physical step remains blocked until Program 90 admits independent data.

Chapter 18

Reproducibility

Executable:

`fin_programs_81_90_asymptotic_operational_completion.py`

Tests:

`test_fin_programs_81_90_asymptotic_operational_completion.py`

Machine-readable results:

`FIN_Programs_81_90_Asymptotic_Operational_Completion_Results.json`

External-data intake template:

`FIN_Programs_81_90_External_Data_Intake_Template.json`

Figures:

`FIN_Programs_81_90_Asymptotic_Operational_Completion_Figures/`

Reproduction:

```
python3 fin_programs_81_90_asymptotic_operational_completion.py python3 -m unittest  
-v test_fin_programs_81_90_asymptotic_operational_completion.py
```

All quantities remain dimensionless unless an external calibration is explicitly supplied. Numerical residuals are verification tolerances, not physical effects.

18.1 Suggested citation

Żuchowski, K. (2026). *FIN Programs 81–90: Asymptotic Spectral Obstructions and Operational Completion Tests* (FIN Research Monograph, Release 10.9; Version 1.0.0) [Preprint].

Archival note

Machine-readable results are stored in `FIN_Programs_81_90_Asymptotic_Operational_Completion_Results.json`. The executable, fifteen tests, six figures, and external-data intake template reproduce the release.

Suggested citation:

Żuchowski, K. (2026). *FIN Programs 81–90: Asymptotic Spectral Obstructions and Operational Completion Tests* (FIN Research Monograph, Release 10.9; Version 1.0.0) [Preprint].